

# Kai Barker

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## EDUCATION

### University of California, Santa Barbara

*Bachelor of Science (B.S.), Statistics and Data Science*

September 2022 - June 2026

Santa Barbara, CA

**Major GPA: 3.90/4.0**

**Activities:** Consult Your Community, Statistics & Data Science Club, Data Science Collaborative, Ski & Snow Club

**Coursework:** Linear Algebra, Optimization, Machine Learning, Stochastic Processes, Time Series, Adv. Statistical Models, Big Data

## SKILLS & CERTIFICATES

**Languages & Tools:** Python (pandas, NumPy, etc.), R (tidyverse, dplyr), Excel, SQL, HTML/CSS, Git, LaTeX, API Integration

**Big Data & Engineering:** Apache Spark, PySpark, GCP (BigQuery, Dataproc, GCS), ETL Pipelines, Parquet Storage, Distributed SQL

**Machine Learning & Statistics:** Scikit-Learn, PyTorch, PCA, XGBoost, Feature Selection (LASSO, NSC), Bootstrap/Permutation Testing

**Visualization & Reporting:** Tableau, Plotly, Seaborn, Matplotlib, ggplot2, Interactive Dashboards, PowerPoint

**Certificates:** JP Morgan Chase Software Engineering (2025), JP Morgan Chase Quantitative Research (2025), HackerRank SQL (2026)

**Interests:** Basketball, Hiking, Backpacking, Rock Climbing, Cooking, Running, Pickleball

## EXPERIENCE

### Research Assistant

September 2024 - Present

*Foundations of Financial Technology Lab*

Santa Barbara, CA

- Quantified extreme wealth concentration in Ethereum (Gini coefficient of 0.96) by building a pipeline to process 20M+ blocks and 100M+ wallets across a 5-year period, showing the top 1% of accounts control 95%+ of Ether supply.
- Improved accuracy of distribution metrics by filtering 65,000+ institutional and artificial accounts to isolate individual holdings.
- Contributed analysis and visualizations to a published blockchain inequality paper with researchers from Yale, Cornell, and UCSB.

### Engagement Manager and Business Analyst

October 2025 - June 2026

*Consult Your Community*

Santa Barbara, CA

- Led a 16-week EdTech market-entry engagement as Engagement Manager, directing a 7-person team across 3 workstreams toward the client's first paid B2B pilot.
- Modeled \$26M in recoverable funding across 115 CA community colleges by building an Excel financial model and Tableau dashboards, communicating findings directly to non-technical clients to drive recommendations.
- Identified \$40,000 in recoverable annual revenue by extracting and merging transactional data from Shopify, Square, and client records, then segmenting corporate accounts by order value, activity, and lifetime value.
- Translated data findings, including that 70% of corporate clients had lapsed within a year, into actionable deliverables: a CRM system, automated email reactivation campaigns, and a buyer-ready SOP for acquisition preparation.

### Data Science Researcher

September 2025 - June 2026

*Department of Statistics and Applied Probability | Advised by Department Chair*

Santa Barbara, CA

- Built an end-to-end Python pipeline to test whether 80,000+ congressional stock trades beat the market, cleaning and reconciling disclosure data to measure abnormal returns against an S&P 500 benchmark.
- Optimized mean-variance portfolios using cvxpy that returned 13.81% annually over an 8-year backtest, outperforming the S&P 500 by +1.07% per year (excluding 2022).
- Improved the reliability of portfolio risk estimates by denoising a 560-asset covariance matrix, applying Random Matrix Theory and Ledoit-Wolf shrinkage to separate true signal from noise.

## PROJECTS

### Gene Therapy Off-Target Effect Analysis

May 2026

- Engineered an end-to-end ETL pipeline on GCP using PySpark, Dataproc, and BigQuery SQL to process a 78,893-gene expression matrix and 705,000+ disease association records across 22 candidate gene therapy treatments for Angelman syndrome research.
- Determined the single safest of 22 candidate treatments by building a distributed validation framework that ran 1,800+ model fits in parallel and flagged statistically significant off-target effects ( $p < 0.001$ ).

### TrailForge

January 2026

- Implemented Strava OAuth 2.0 to ingest user activity data (distance, elevation, heart rate zones) and construct structured fitness profiles as input to a Gemini LLM pipeline for personalized trail recommendations.
- Resolved LLM URL hallucination failures by pivoting to a Google SerpAPI integration for reliable AllTrails link retrieval, integrating three external APIs into a deployed web app within a 24-hour hackathon window.

### NeurIPS Ariel Data Challenge

July 2025

- Designed a signal processing pipeline to calibrate 200+ GB of raw telescope data across 14,000 files and 1,100+ exoplanets, correcting detector noise and instrument distortion to recover usable atmospheric signal for the European Space Agency's Ariel mission.
- Constructed a weighted ensemble of Gaussian Process Regression, XGBoost, and Ridge models to predict atmospheric composition across 283 wavelength channels, tuning uncertainty estimates via grid search to optimize Gaussian Log Likelihood.